raining down on their detectors. The most sensitive of the world's cosmic ray detectors, the Pierre Auger Observatory—with an observing area the size of Rhode Island—is now being built on a vast stretch of land in western Argentina. Shapere and Feng estimate that if all of the extra dimensions are as large as 10^{-14} meters, then after a year's worth of data collection, the Auger detector will see the characteristic particle debris from about a dozen tiny black holes produced in the upper atmosphere. If such black hole signatures are not found, the experiment will conclude that extra dimensions are smaller. Finding the remains of black holes produced in cosmic ray collisions is certainly a long shot, but success would open the first experimental window on extra dimensions, black holes, string theory, and quantum gravity.

Beyond black hole production, there is another, accelerator-based way that researchers will be looking for extra dimensions during the next decade. The idea is a sophisticated variant on the "space-between-the-cushions" explanation for the loose coins missing from your pocket.

A central principle of physics is conservation of energy. Energy can manifest itself in many forms—the kinetic energy of a ball's motion as it flies off a baseball bat, gravitational potential energy as the ball flies upward, sound and heat energy when the ball hits the ground and excites all sorts of vibrational motion, the mass energy that's locked inside the ball itself, and so on—but when all carriers of energy have been accounted for, the amount with which you end always equals the amount with which you began.[7] To date, no experiment contradicts this law of perfect energy balance.

But depending on the precise size of the hypothesized extra dimensions, high-energy experiments to be carried out at the newly upgraded facility at Fermilab and at the Large Hadron Collider may reveal processes that appear to violate energy conservation: the energy at the end of a collision may be less than the energy at the beginning. The reason is that, much like your missing coins, energy (carried by gravitons) can seep into the cracks—the tiny additional space—provided by the extra dimensions and hence be inadvertently overlooked in the energy accounting calculation. The possibility of such a "missing energy signal" provides yet another means for establishing that the fabric of the cosmos has complexity well beyond what we can see directly.